

21 UCSC Genome Browser Command-line Tools

Learning Objectives

By the end of the session, students will be able to:

- Understand data formats used for storing large genomic datasets
- Extract data from these, especially via the UCSC Genome Browser

Some data formats

You already know:

- VCF (and BCF, the binary version)
 - Stores variant-level info, including genotypes
- BED
 - Not the PLINK bed
 - Describes regions of the genome
 - Flexible (3-12 columns)

Perhaps new to you:

- GFF/GTF
- bigBed
- bigWig

VCF refresher

The following few slides on VCF were stolen from Ryan Minster!

A VCF Example

```
##fileformat=VCFv4.2
##fileDate=20090805
##contig=<ID=chr1>
##contig=<ID=chr2>
##contig=<ID=chr3>
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=AC,Number=A,Type=Integer,Description="Alternate Allele Counts">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total Number Allele Counts">
##INFO=<ID=GC,Number=G,Type=Integer,Description="Genotype Counts">
##INFO=<ID=GN,Number=1,Type=Integer,Description="Total Number of Genotypes">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 151">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
#CHROM POS    ID REF ALT QUAL FILTER INFO                FORMAT S1 S2 S3
chr1 46634552 . G  A  .  .   NS=3;AC=3;AN=6;GC=1,1,1;GN=3 GT  0/0 0/1 1/1
chr1 47832477 . T  A  .  .   NS=3;AC=5;AN=6;GC=0,1,2;GN=3 GT  1/1 1/1 0/1
chr2 132488276 . A  .  .  .   NS=2;AC=0;AN=4;GC=2,0,0;GN=2 GT  ./ 0/0 0/0
chr3 4387773  rs1 G  GT  .  .   NS=3;AC=1;AN=6;GC=2,1,0;GN=3;DB GT  0/0 0/1 0/0
```

Data Lines Fields 1–8

1	2	3	4	5	6	7	8
#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO
chr1	695744	.	C	T	.	PASS	AA=.;DP=115
chr1	713682	rs11804171	T	A	.	PASS	AA=.;DP=137;HM2
chr1	726158	.	A	T	.	PASS	AA=.;DP=73
chr1	742429	rs3094315	G	A	.	PASS	AA=g;DP=96;HM2

INFO Field

The **INFO** field contains a semicolon (;)-delimited list of information about the variant.

```
#CHROM POS ID REF ALT QUAL FILTER INFO
```

```
chr3 4387773 rs1 G T . . NS=3;AC=1;AN=6;GC=2,1,0;GN=3;DB
```

```
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
```

```
##INFO=<ID=AC,Number=A,Type=Integer,Description="Alternate Allele Counts">
```

```
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total Number Allele Counts">
```

```
##INFO=<ID=GC,Number=G,Type=Integer,Description="Genotype Counts">
```

```
##INFO=<ID=GN,Number=1,Type=Integer,Description="Total Number of Genotypes">
```

```
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 151">
```

Data Lines Fields 9, 10, ...

3	4	5	9	10	11	12
ID	REF	ALT	FORMAT	001	002	003
.	C	T	GT:GQ:DP	0/1:100:20	0 0:16:13	0 0:100:82
rs11804171	T	A	GT:GQ:DP	0/1:33:30	0 1:54:14	0 0:100:43
.	A	T	GT:GQ:DP	0/1:100:26	0 1:100:20	0 0:72:25
rs3094315	G	A	GT:GQ:DP	1/1:69:23	1 0:98:22	1 1:100:47

##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">

##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">

##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">

GTF/GFF

- For storing genome annotations (not variants)
 - Genes
 - Transcripts
 - Exons
 - Regulatory elements
 - Etc.
- Think “BED + more structure about biological features + relationships”
- Ultimate source of data for many Genome Browser gene annotation tracks

GTF/GFF columns

Columns (1-8 are the same)

- chrom (chromosome)
- source (annotation source; e.g., HAVANA, Ensembl)
- feature (gene, transcript, exon, CDS, etc.)
- start/end (coordinates)
- score (often .)
- strand (+ or -)
- frame (reading frame [0/1/2 or .])

Column 9 (details)

- Can encode hierarchical relationships between rows of the dataset
 - Which exon belongs to which transcript
 - Which transcript belongs to which gene
 - Etc.
- Key-value pairs (like INFO in a VCF), e.g.:
 - GTF: gene_id "ENSG..."; transcript_id "ENST...";
 - GFF: ID=gene:ENSG...;Name=DDX11L2;biotype=lncRNA;

GTF/GFF

```
> curl -L https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_45/gencode.v45.annotation.gtf.gz | gunzip | head
% Total    % Received % Xferd  Average Speed   Time    Time     Current
                                 Dload  Upload  Total   Spent    Left     Speed
  0     0    0     0    0     0      0      0      0     0  0:00:00  0:00:00  0:00:00
##provider: GENCODE
##contact: gencode-help@ebi.ac.uk
##format: gtf
##date: 2023-09-19
chr1  HAVANA  gene  11869  14409  .  +  .  gene_id "ENSG00000290825.1"; gene_type "lncRNA"; gene_name "DDX11L2"; level 2; tag "overlaps_pseudogene";
chr1  HAVANA  transcript  11869  14409  .  +  .  gene_id "ENSG00000290825.1"; transcript_id "ENST00000456328.2"; gene_type "lncRNA"; gene_name "DDX11L2"; transcript_type "lncRNA"; transcript_name "DDX11L2-202"; level 2; transcript_support_level "1"; tag "basic"; tag "Ensembl_canonical"; havana_transcript "OTTHUMT00000362751.1";
chr1  HAVANA  exon  11869  12227  .  +  .  gene_id "ENSG00000290825.1"; transcript_id "ENST00000456328.2"; gene_type "lncRNA"; gene_name "DDX11L2"; transcript_type "lncRNA"; transcript_name "DDX11L2-202"; exon_number 1; exon_id "ENSE00002234944.1"; level 2; transcript_support_level "1"; tag "basic"; tag "Ensembl_canonical"; havana_transcript "OTTHUMT00000362751.1";
chr1  HAVANA  exon  12613  12721  .  +  .  gene_id "ENSG00000290825.1"; transcript_id "ENST00000456328.2"; gene_type "lncRNA"; gene_name "DDX11L2"; transcript_type "lncRNA"; transcript_name "DDX11L2-202"; exon_number 2; exon_id "ENSE00003582793.1"; level 2; transcript_support_level "1"; tag "basic"; tag "Ensembl_canonical"; havana_transcript "OTTHUMT00000362751.1";
chr1  HAVANA  exon  13221  14409  .  +  .  gene_id "ENSG00000290825.1"; transcript_id "ENST00000456328.2"; gene_type "lncRNA"; gene_name "DDX11L2"; transcript_type "lncRNA"; transcript_name "DDX11L2-202"; exon_number 3; exon_id "ENSE00002312635.1"; level 2; transcript_support_level "1"; tag "basic"; tag "Ensembl_canonical"; havana_transcript "OTTHUMT00000362751.1";
0.47.4M  0  63748  0  0 133k  0 0:06:04 --:--: 0:06:04 133k
curl: (23) Failure writing output to destination, passed 8000 returned 0
```

```
[> curl -Ls https://ftp.ensembl.org/pub/release-110/gff3/homo_sapiens/Homo_sapiens.GRCh38.110.gff3.gz | zcat | grep -v "^#" | head
1      GRCh38  chromosome      1      248956422      .      .      .      ID=chromosome:1;Alias=CM000663.2,chr1,NC_000001.11
1      .      biological_region  10469  11240  1.3e+03  .      .      external_name=oe %3D 0.79;logic_name=cpg
1      .      biological_region  10650  10657  0.999  +      .      logic_name=eponine
1      .      biological_region  10655  10657  0.999  -      .      logic_name=eponine
1      .      biological_region  10678  10687  0.999  +      .      logic_name=eponine
1      .      biological_region  10681  10688  0.999  -      .      logic_name=eponine
1      .      biological_region  10707  10716  0.999  +      .      logic_name=eponine
1      .      biological_region  10708  10718  0.999  -      .      logic_name=eponine
1      .      biological_region  10735  10747  0.999  -      .      logic_name=eponine
1      .      biological_region  10737  10744  0.999  +      .      logic_name=eponine
```

bigBed, etc.

- bigBed = compressed, indexed version of BED for large datasets
 - Binary format
 - Internally indexed
 - Fast, remote queries
 - Used by UCSC genome browser to underlie tracks
- Use UCSC utilities to access
 - bigBedToBed (most important: query a region and convert to BED)
 - bigBedSummary (summary over a region)
 - bigBedInfo (inspect file at high level)
 - bedToBigBed (convert in the other direction)

Downloading files with wget

wget

- Command-line tool to download files from the web
- Non-interactive (so good for scripts/pipelines)
- Example: `wget -O myfile.vcf.gz https://example.com/file.vcf.gz`
- Can resume interrupted downloads
 - use the `-c` option
 - very helpful for large files
- Often installed by default in Linux (often not for Mac)

Transferring data with curl

curl

- Command-line tool to transfer data from URLs
- More flexible than wget
- Can stream/pipe output instead of saving to file
- Examples:
 - `curl -o myfile.vcf.gz https://example.com/file.vcf.gz`
 - `curl https://example.com/file.vcf.gz | gunzip | head`

Querying genomic regions with tabix

tabix

- Command-line tool to query genomic regions from large, indexed files
- Works locally or remotely
- Avoid downloading or scanning whole files
- Works on: VCF, BED, GFF
- Requires
 - bgzip compression (.gz, not regular gzip)
 - index file (.tbi)
- Very fast because of the index: jumps directly to region (doesn't scan the whole file)
- Example: `tabix file.vcf.gz chr1:1000000-2000000`
- You can create an index, e.g.:
 - `bgzip file.vcf`
 - `tabix -p vcf file.vcf.gz`

Querying VCF/BCF with bcftools

bcftools

- Toolkit just for VCF/BCF
- Like tabix
 - Region queries
 - Remote or local
 - Uses indexing
- Unlike tabix
 - Understands full VCF structure
 - Filter variants
 - Filtering genotypes
 - Extracting specific fields

Swiss-Army knife for genomic intervals: bedtools

- Perform set operations on genomic intervals in multiple files
 - BED
 - GFF/GTF
 - VCF
- Common operations:
 - Intersect (overlap between datasets)
 - Merge (combine overlapping regions)
 - Closest (nearest feature)
 - Coverage (how much overlap/depth)
- Doesn't use indices like tabix
- Scans through files

UCSC utilities

UCSC provides many command-line tools

- Access UCSC-specific file formats
- Extract data from large datasets, remotely
- Convert between formats
- Often use internal indices (no index file required)

Summary

Tool	Use
wget	“I want to download a file from the web”
curl	“I want to pipe web data directly into other commands”
tabix	“I want info on some regions from a huge file”
bcftools	“I want to query/filter VCF/BCF data”
bedtools	“I already have genomic data; now I want to compare/manipulate it”
bigBedtoBed, etc.	“I want to extract data from UCSC-hosted tracks or binary formats”

Example bigBedToBed Code

```
bigBedToBed -header -chrom=chr19 \  
-start=44905795 -end=44909393 \  
http://hgdownload.soe.ucsc.edu/gbdb/hg38/mane/mane.1.0.bb \  
output.bed
```

This extracts rows from MANE select v1.0 (table: mane) within 10 kb of the peak SNP

Then use Unix tools or import the file into R to manipulate.

It is not as versatile as SQL for pulling down data, but it suffices.

Example bigBedToBed Code

```
bigBedToBed -header -chrom=chr19 \  
-start=44898822 -end=44918822 \  
http://hgdownload.soe.ucsc.edu/gbdb/hg38/mane/mane.1.0.bb \  
stdout | cut -f1-3,6,19 | column -s $'\t' -t
```

This extracts rows from MANE select v1.0 (table: mane) within 10 kb of the peak SNP, then processes the output so that you see just columns 1–3, 6, and 19

#chrom	chromStart	chromEnd	strand	geneName2
chr19	44891253	44903689	+	TOMM40
chr19	44905795	44909393	+	APOE
chr19	44914607	44919346	+	APOC1

Example bigBedToBed Code

```
bigBedToBed -header -chrom=chr19 \  
-start=44908797 -end=44908847 \  
http://hgdownload.soe.ucsc.edu/gbdb/hg38/bbi/clinvar/clinvarMain.bb \  
stdout | cut -f13 | column -s $'\t' -t
```

These are all the ClinVar variants from (clinvarMain) within 50 bp of the peak SNP

origName

```
478850|NM_000041.4(APOE):c.508G>C (p.Ala170Pro)  
694741|NM_000041.4(APOE):c.526= (p.Arg176=)  
17848|NM_000041.2(APOE):c.526C>T (p.Arg176Cys)  
1341575|NM_000041.4(APOE):c.548G>C (p.Gly183Ala)
```

Example bigBedToBed Code

```
bigBedToBed -header -maxItems=4 -chrom=chr19 \  
-start=44908797 -end=44908847 \  
http://hgdownload.soe.ucsc.edu/gbdb/hg38/snp/dbSnp153.bb \  
stdout | cut -f4,5,7 | column -s $'\t' -t
```

*First four SNPs in the MANE APOE transcript from dbSNP 153
(table: dbSnp153)*

name	ref	alts
rs515726148	CTCCTCC	CTCC,
rs867594573	C	A,T,
rs376170967	G	A,
rs1300930146	G	T,